Eigenvalues and diagonalization

Linear Algebra · Lecture of 11 November 2026

Goal: for a square matrix A, find the vectors $x \neq 0$ that A only stretches, $Ax = \lambda x$. When $\mathbb{R}^n$ has a basis of such vectors, $A = P\Lambda P^{-1}$ with Λ diagonal, and powers and inverses of A become easy.

Use: trace and determinant as the sum and the product of the eigenvalues; the eigenvalues of $X'X$ and XX'; from the next lecture on, covariance matrices and their principal axes.

Last time (before the midterm, Chapters 1–6): matrices, linear systems, subspaces, rank, determinants and inner products. The results used today are listed as Facts F1–F10; each is proved in Chapters 1–6 of the textbook. Throughout, $I = I_n$, and $e_1, \dots, e_n$ are the columns of I_n.

Facts from Chapters 1–6

Fact F1 (matrix arithmetic)**.** For matrices of compatible sizes and numbers c: $(AB)C = A(BC)$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$, $c(AB) = (cA)B = A(cB)$, and $I_m A = A = AI_n$ for $A \in \mathbb{R}^{m\times n}$. Matrices of the same size are added and multiplied by numbers entry by entry, and these operations obey the rules of vector arithmetic: $A + B = B + A$, $c(A + B) = cA + cB$, $(c + d)A = cA + dA$ and $(cd)A = c(dA)$. Block matrices with the same partition are added and multiplied by numbers block by block.

Fact F2 (products by columns)**.** Let $A \in \mathbb{R}^{m\times n}$ have the columns $a_1, \dots, a_n$; $\operatorname{col}_j(A) = a_j$ denotes column j. For $x = (x_1, \dots, x_n)'$, $Ax = x_1a_1 + \cdots + x_na_n$; in particular $Ae_j = a_j$. For $B \in \mathbb{R}^{n\times p}$, $\operatorname{col}_j(AB) = A \operatorname{col}_j(B)$. Two matrices of the same size are equal if and only if their columns are.

Fact F3 (transpose, trace, ones)**.** $(A')' = A$, $(A + B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$ and $I' = I$. The **trace** of $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is $\operatorname{tr} A = a_{11} + \cdots + a_{nn}$. $\mathbf{1}_n = (1, \dots, 1)' \in \mathbb{R}^n$, $\mathbf{1}_n'\mathbf{1}_n = n$, and $J_n = \mathbf{1}_n\mathbf{1}_n'$ is the $n \times n$ matrix of ones. A vector $x \in \mathbb{R}^n$ is **centered** if $\mathbf{1}_n'x = 0$.

Fact F4 (invertible matrices)**.** For $A \in \mathbb{R}^{n\times n}$ the following are equivalent: A is invertible; $Ax = 0$ only for $x = 0$; $\operatorname{rank} A = n$; the columns of A are linearly independent; the columns of A form a basis of $\mathbb{R}^n$. If A and B are invertible, so are AB and A^{-1}, and $(AB)^{-1} = B^{-1}A^{-1}$, $(A^{-1})^{-1} = A$.

Fact F5 (independence, bases, dimension)**.** If $cx = 0$ with $x \neq 0$, then $c = 0$. A list that contains the zero vector is linearly dependent. $\dim \mathbb{R}^n = n$. Let W be a subspace of dimension d. If W contains g linearly independent vectors, then $g \leqslant d$; any d linearly independent vectors of W form a basis of W. Every linearly independent list in $\mathbb{R}^n$ can be extended to a basis of $\mathbb{R}^n$.

Fact F6 (null space and rank)**.** For $A \in \mathbb{R}^{m\times n}$, the null space $\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$ is a subspace of $\mathbb{R}^n$, and $\dim \mathcal{N}(A) = n - \operatorname{rank} A$. The rank is the number of nonzero rows of an echelon form of A, and the special solutions read off from the reduced row echelon form form a basis of $\mathcal{N}(A)$. Row operations do not change the null space: if R is obtained from A by row operations, then $\mathcal{N}(R) = \mathcal{N}(A)$. $\operatorname{rank}(AB) \leqslant \operatorname{rank} A$, $\operatorname{rank}(AB) \leqslant \operatorname{rank} B$, and $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for invertible P, Q. A matrix whose g columns are linearly independent has rank g.

Fact F7 (determinants)**.** For $A, B \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}$: A is invertible if and only if $\det A \neq 0$; $\det(AB) = \det A \det B$; $\det(cA) = c^n \det A$; $\det A' = \det A$; $\det I = 1$. If A is upper or lower triangular, $\det A = a_{11}a_{22} \cdots a_{nn}$. Adding a multiple of one row to another row does not change $\det A$, and multiplying one row by c multiplies $\det A$ by c; the same holds for columns.

Fact F8 (cofactor expansion)**.** For $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ let $A(i|j)$ be A without row i and column j. For every row i, $\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det A(i|j)$; for every column j, $\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det A(i|j)$.

Fact F9 (block determinants)**.** Let $M = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$ with $P \in \mathbb{R}^{p\times p}$ and $S \in \mathbb{R}^{q\times q}$. If $Q = 0$ or $R = 0$, then $\det M = \det P \det S$. If P is invertible, then $\det M = \det P \det(S - RP^{-1}Q)$; if S is invertible, then $\det M = \det S \det(P - QS^{-1}R)$.

Fact F10 (span)**.** The **span** $\operatorname{span}\{x_1, \dots, x_k\}$ of vectors $x_1, \dots, x_k \in \mathbb{R}^n$ is the set of all linear combinations $c_1x_1 + \cdots + c_kx_k$; it is a subspace of $\mathbb{R}^n$.

1. Eigenvalues and eigenvectors

Definition 1. Let $A \in \mathbb{R}^{n\times n}$. A real number λ is an **eigenvalue** of A if $Ax = \lambda x$ for some $x \in \mathbb{R}^n$ with $x \neq 0$; such a vector x is an **eigenvector** of A **belonging to** λ. Since $Ax = \lambda x$ if and only if $(A - \lambda I)x = 0$, the set

$$\mathcal{N}(A - \lambda I) = \{x \in \mathbb{R}^n : Ax = \lambda x\}$$

is called the **eigenspace** of A for λ.

The eigenspace $\mathcal{N}(A - \lambda I)$ is a subspace of $\mathbb{R}^n$ (Fact F6), and its nonzero vectors are exactly the eigenvectors belonging to λ. So a nonzero multiple of an eigenvector, and a nonzero sum of eigenvectors belonging to the same λ, are again eigenvectors belonging to λ. An eigenvector belongs to only one eigenvalue: if $Ax = \lambda_1 x$ and $Ax = \lambda_2 x$, then $(\lambda_1 - \lambda_2)x = 0$ with $x \neq 0$, so $\lambda_1 = \lambda_2$ (Fact F5).

Theorem 2. Let $A \in \mathbb{R}^{n\times n}$ and $\lambda \in \mathbb{R}$. The following are equivalent:

(i) λ is an eigenvalue of A;

(ii) $A - \lambda I$ is not invertible;

(iii) $\det(\lambda I - A) = 0$;

(iv) $\operatorname{rank}(A - \lambda I) < n$.

Proof.
- (i)⇔(ii): λ is an eigenvalue if and only if $(A - \lambda I)x = 0$ has a solution $x \neq 0$, that is, if and only if $A - \lambda I$ is not invertible (Fact F4).
- (ii)⇔(iii): $\det(\lambda I - A) = \det\big((-1)(A - \lambda I)\big) = (-1)^n \det(A - \lambda I)$, as $\det(cB) = c^n \det B$; and $A - \lambda I$ is invertible if and only if $\det(A - \lambda I) \neq 0$ (Fact F7).
- (ii)⇔(iv): $A - \lambda I$ is invertible if and only if $\operatorname{rank}(A - \lambda I) = n$ (Fact F4). □

So the eigenvalues of A in the sense of Definition 1 are the real solutions λ of $\det(\lambda I - A) = 0$, and the eigenvectors belonging to λ are the nonzero solutions of $(A - \lambda I)x = 0$, found by row reduction.

2. Polynomials

Complex numbers. A **complex number** is $z = a + b\mathrm{i}$ with $a, b \in \mathbb{R}$, where $\mathrm{i}^2 = -1$; $\mathbb{C}$ is the set of complex numbers, and a real number a is the complex number $a + 0\mathrm{i}$. Sums and products are $(a + b\mathrm{i}) + (c + d\mathrm{i}) = (a + c) + (b + d)\mathrm{i}$ and $(a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}$; they obey the rules of arithmetic of real numbers (commutative, associative and distributive laws), as one checks by multiplying out both sides. The **conjugate** of $z = a + b\mathrm{i}$ is $\bar{z} = a - b\mathrm{i}$, and $z\bar{z} = a^2 + b^2 > 0$ for $z \neq 0$. Hence every $z \neq 0$ has the inverse $z^{-1} = \bar{z}/(a^2 + b^2)$, and $zw = 0$ only if $z = 0$ or $w = 0$ (if $z \neq 0$, then $w = z^{-1}zw = 0$): a product of complex numbers is 0 only if one of the factors is 0.

Definition 3. A **polynomial** (in the variable t) is an expression $f = c_0 + c_1 t + \cdots + c_m t^m$ with **coefficients** $c_0, \ldots, c_m \in \mathbb{C}$; we put $c_k = 0$ for $k > m$, and f equals $g = d_0 + d_1 t + \cdots$ if $c_k = d_k$ for every $k \geqslant 0$. We write $f(t)$ for f when the variable is to be shown.

- f is **real** if every $c_k \in \mathbb{R}$; the **zero polynomial** 0 has all coefficients 0. For $f \neq 0$, the **degree** $\deg f$ is the largest k with $c_k \neq 0$, and $c_{\deg f}$ is the **leading coefficient**; f is **monic** if its leading coefficient is 1. c_0 is the **constant term**, and a polynomial with $c_k = 0$ for all $k \geqslant 1$ is written as the number c_0.
- The sum $f + g$, the multiple af ($a \in \mathbb{C}$) and the product fg have the coefficients $c_k + d_k$, ac_k and $\sum_{i+j=k} c_i d_j = c_0 d_k + c_1 d_{k-1} + \cdots + c_k d_0$; thus $(ct^i)(dt^j) = cd\, t^{i+j}$. These operations obey the rules of arithmetic of numbers ($fg = gf$, $(fg)h = f(gh)$, $f(g+h) = fg + fh$, and so on), because these hold coefficient by coefficient; for instance, with $h = \sum_l h_l t^l$, the coefficient of t^k in $(fg)h$ and in $f(gh)$ is $\sum_{i+j+l=k} c_i d_j h_l$.
- The **value** of f at $z \in \mathbb{C}$ is $f(z) = c_0 + c_1 z + \cdots + c_m z^m$, and z is a **root** of f if $f(z) = 0$. For a real polynomial f and $A \in \mathbb{R}^{n\times n}$, $f(A) = c_0 I + c_1 A + \cdots + c_m A^m$, where $A^0 = I$ and $A^{k+1} = A^k A$; for instance, for $f = t - c$, $f(A) = A - cI$.

Proposition 4. Let f, g, h be polynomials.

(a) If $f \neq 0$ and $g \neq 0$, then $fg \neq 0$, $\deg(fg) = \deg f + \deg g$, and the leading coefficient of fg is the product of those of f and g. If $f \neq 0$ and $fg = fh$, then $g = h$.

(b) $(af + g)(z) = af(z) + g(z)$ and $(fg)(z) = f(z)g(z)$ for $a, z \in \mathbb{C}$.

(c) (**Factor theorem**) For $c \in \mathbb{C}$, $f(c) = 0$ if and only if $f = (t - c)q$ for some polynomial q.

(d) A nonzero polynomial of degree m has at most m roots. Consequently, if $f(z) = g(z)$ for infinitely many z (for instance, for all real z, or for all real $z \neq 0$), then $f = g$.

(e) For $\mu_1, \ldots, \mu_n \in \mathbb{C}$ ($n \geqslant 1$), $(t - \mu_1)\cdots(t - \mu_n) = t^n - (\mu_1 + \cdots + \mu_n)t^{n-1} + r$, where $r = 0$ or $\deg r \leqslant n - 2$; and the constant term of this product is $(-1)^n \mu_1 \cdots \mu_n$.

Proof. Write $f = \sum_k c_k t^k$ and $g = \sum_k d_k t^k$.

(a)

- Let $\deg f = m$ and $\deg g = l$. A term $c_i d_{m+l+j-i}$ ($j \geqslant 0$) of the coefficient of t^{m+l+j} in fg can be nonzero only if $i \leqslant m$ and $m + l + j - i \leqslant l$, that is, $m + j \leqslant i \leqslant m$; so $j = 0$ and $i = m$. Hence the coefficient of t^{m+l} is $c_m d_l$, which is $\neq 0$ as $c_m \neq 0$ and $d_l \neq 0$ (a product of complex numbers is 0 only if a factor is 0); all higher coefficients are 0.
- If $fg = fh$, then $f(g - h) = 0$; if $g \neq h$, the first item would give $f(g - h) \neq 0$ ✘ So $g = h$.

(b)

- $af + g$ has the coefficients $ac_k + d_k$, so $(af + g)(z) = \sum_k (ac_k + d_k)z^k = af(z) + g(z)$.
- By the product rule of Definition 3 and the distributive laws, $(fg)(z) = \sum_k \sum_{i+j=k} c_i d_j z^{i+j} = \sum_{i,j} (c_i z^i)(d_j z^j) = f(z)g(z)$.

(c)

- First, $f = (t - c)q + r$ for a polynomial q and a number r, by induction on the degree: if $f = 0$ or $\deg f = 0$, take $q = 0$ and $r = f$; if $\deg f = m \geqslant 1$ and a is the leading coefficient, then $f - at^{m-1}(t - c)$ has coefficient 0 at t^m, so it is 0 or of degree less than m; by the induction hypothesis it equals $(t - c)q_1 + r$, and $f = (t - c)(q_1 + at^{m-1}) + r$.
- By (b), $f(c) = (c - c)q(c) + r = r$. So if $f(c) = 0$, then $r = 0$ and $f = (t - c)q$. Conversely, if $f = (t - c)q$, then $f(c) = (c - c)q(c) = 0$ by (b).

(d)

- Induction on m. If $m = 0$, f is a nonzero number and has no root. Let $m \geqslant 1$, and let f have a root a (otherwise there is nothing to prove). By (c), $f = (t-a)q$; here $q \neq 0$, as $f \neq 0$, so $\deg q = m-1$ by (a).
- If $f(b) = 0$, then $(b-a)q(b) = 0$ by (b), so $b = a$ or $q(b) = 0$ (a product is 0 only if a factor is 0). By the induction hypothesis q has at most $m-1$ roots, so f has at most m.
- If $f \neq g$, then $f - g$ is a nonzero polynomial, and every z with $f(z) = g(z)$ is a root of it (by (b)); so there are only finitely many such z.

(e)

- Induction on n; for $n = 1$, $t - \mu_1$ has this form with $r = 0$.
- If $(t-\mu_1)\cdots(t-\mu_n) = t^n - s_n t^{n-1} + r_n$ with $s_n = \mu_1 + \cdots + \mu_n$, then multiplying by $t - \mu_{n+1}$ gives $t^{n+1} - (s_n + \mu_{n+1})t^n + r_{n+1}$ with $r_{n+1} = s_n\mu_{n+1}t^{n-1} + r_n(t - \mu_{n+1})$, which is 0 or of degree at most $n-1$ by (a).
- The constant term of a polynomial is its value at 0, here $(0-\mu_1)\cdots(0-\mu_n) = (-1)^n\mu_1\cdots\mu_n$ by (b). □

3. The characteristic polynomial

Lemma 5. For every $t \in \mathbb{R}$ let $M(t) \in \mathbb{R}^{m\times m}$ have the (i,j) entry $\alpha_{ij} + \beta_{ij}t$, where the numbers $\alpha_{ij}, \beta_{ij} \in \mathbb{R}$ do not depend on t. If at most d rows of $M(t)$ contain an entry with $\beta_{ij} \neq 0$, then there is a real polynomial q, equal to 0 or of degree at most d, with $\det M(t) = q(t)$ for every $t \in \mathbb{R}$.

Proof. Induction on m.

- $m = 1$: $\det M(t) = \alpha_{11} + \beta_{11}t$, as a 1×1 matrix is triangular (Fact F7); take $q = \alpha_{11} + \beta_{11}t$, which is a number if $d = 0$, as then $\beta_{11} = 0$.
- Let $m \geqslant 2$. If no entry has $\beta_{ij} \neq 0$, then $\det M(t)$ is a number that does not depend on t; take q to be this number. Otherwise $d \geqslant 1$; let row i contain an entry with $\beta_{ij} \neq 0$, and expand along row i (Fact F8):
$$\det M(t) = \sum_{j=1}^{m} (-1)^{i+j}(\alpha_{ij} + \beta_{ij}t)\det M(t)(i|j)$$
- Each $M(t)(i|j) \in \mathbb{R}^{(m-1)\times(m-1)}$ has entries of the same form, and its rows are the rows of $M(t)$ other than row i, each without its jth entry; so at most $d-1$ of them contain an entry with $\beta \neq 0$. By the induction hypothesis, $\det M(t)(i|j) = q_j(t)$ for every t, with $q_j = 0$ or $\deg q_j \leqslant d-1$.
- So $\det M(t) = q(t)$ for $q = \sum_j (-1)^{i+j}(\alpha_{ij} + \beta_{ij}t)q_j$ (Proposition 4(b)), and each term of this sum is 0 or of degree at most d (Proposition 4(a)); hence so is q. □

Theorem 6. Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$.

(a) There is exactly one polynomial p_A with $p_A(c) = \det(cI - A)$ for every $c \in \mathbb{R}$; it is real. It is the **characteristic polynomial** of A, written $p_A(t) = \det(tI - A)$, and $p_A(t) = 0$ is the **characteristic equation** of A.

(b) p_A is monic of degree n, its coefficient of t^{n-1} is $-\operatorname{tr} A$, and its constant term is $(-1)^n \det A$:
$$p_A(t) = t^n - (\operatorname{tr} A)\, t^{n-1} + \cdots + (-1)^n \det A$$
(for $n = 1$, $p_A(t) = t - a_{11}$, and the last two descriptions coincide). For $n = 2$, $p_A(t) = t^2 - (\operatorname{tr} A)t + \det A$.

(c) A real number λ is an eigenvalue of A in the sense of Definition 1 if and only if $p_A(\lambda) = 0$. Hence A has at most n such eigenvalues.

Proof. (a) The (i, j) entry of $cI - A$ is $\delta_{ij}c - a_{ij}$ ($\delta_{ii} = 1$, $\delta_{ij} = 0$ for $i \neq j$), of the form of Lemma 5 with $d = n$. So a real polynomial q with $q(c) = \det(cI - A)$ for every real c exists. Two such polynomials agree at every real c, so they are equal (Proposition 4(d)).

(b) Induction on n. For $n = 1$, $\det(cI - A) = c - a_{11}$ (Fact F7), so $p_A = t - a_{11}$, and $-a_{11} = -\operatorname{tr} A = (-1)^1 \det A$. Let $n \geqslant 2$ and $c \in \mathbb{R}$. The first row of $cI - A$ is $(c - a_{11}, -a_{12}, \ldots, -a_{1n})$, and $(cI - A)(1|1) = cI_{n-1} - A(1|1)$. Expanding along row 1 (Fact F8),

$$\det(cI - A) = (c - a_{11})\, p_{A(1|1)}(c) + \sum_{j=2}^{n} (-1)^{1+j}(-a_{1j}) \det\Big((cI - A)(1|j)\Big)$$

- For $j \geqslant 2$, the rows of $(cI - A)(1|j)$ are rows $2, \ldots, n$ of $cI - A$ without their jth entry. Row i of $cI - A$ contains c only in column i, so the row that comes from row j contains no c, and at most $n - 2$ rows contain c. By Lemma 5, $\det\Big((cI - A)(1|j)\Big) = q_j(c)$ with $q_j = 0$ or $\deg q_j \leqslant n - 2$.
- By the induction hypothesis, $p_{A(1|1)} = t^{n-1} - \operatorname{tr}\Big(A(1|1)\Big)t^{n-2} + r$ with $r = 0$ or $\deg r \leqslant n - 3$ ($r = 0$ if $n = 2$). Multiplying out, $(t - a_{11})\, p_{A(1|1)} = t^n - \Big(a_{11} + \operatorname{tr} A(1|1)\Big)t^{n-1} + s$ with $s = 0$ or $\deg s \leqslant n - 2$, and $a_{11} + \operatorname{tr} A(1|1) = \operatorname{tr} A$.
- So $\det(cI - A)$ is the value at c of $t^n - (\operatorname{tr} A)t^{n-1} + s + \sum_{j \geqslant 2}(-1)^{1+j}(-a_{1j})q_j$ (Proposition 4(b)), for every real c. By (a) this polynomial is p_A: it is monic of degree n, with coefficient $-\operatorname{tr} A$ of t^{n-1}.
- The constant term is the value at 0: $p_A(0) = \det(-A) = (-1)^n \det A$, as $\det(cB) = c^n \det B$ (Fact F7). For $n = 2$ these are all three coefficients.

(c) By Theorem 2, $\lambda \in \mathbb{R}$ is an eigenvalue if and only if $p_A(\lambda) = \det(\lambda I - A) = 0$. By (b) and Proposition 4(d), p_A has at most n roots. □

In examples, $\det(tI - A)$ is computed for each real t by row and column operations and cofactor expansion (Facts F7 and F8). If this gives $\det(tI - A) = q(t)$ for a polynomial q and every real t, then $p_A = q$, by Theorem 6(a). Many books call $\det(A - tI) = (-1)^n p_A(t)$ the characteristic polynomial; it has the same roots.

Example 7. Find the eigenvalues of $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & 3 \end{pmatrix}$ and a basis of each eigenspace.

Solution.

- For $t \in \mathbb{R}$, expanding along row 1 (Fact F8),

$$\det(tI - A) = \det \begin{pmatrix} t-1 & -2 & -1 \\ -2 & t & 2 \\ 1 & -2 & t-3 \end{pmatrix} = (t-1)\Big(t(t-3)+4\Big) + 2\Big(-2(t-3)-2\Big) - (4-t)$$
$$= (t^3 - 4t^2 + 7t - 4) + (-4t + 8) + (t - 4) = t^3 - 4t^2 + 4t = t(t-2)^2$$

  So $p_A(t) = t(t-2)^2$, and the eigenvalues are 0 and 2 (Theorem 6(c)).

- $\lambda = 0$: row-reduce $A - 0I = A$.

$$A \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow -\frac{1}{4}R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The free unknown is x_3; the special solution $(1,-1,1)'$ is a basis of $\mathcal{N}(A)$ (Fact F6).

- $\lambda = 2$: row-reduce $A - 2I$.

$$\begin{pmatrix} -1 & 2 & 1 \\ 2 & -2 & -2 \\ -1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} -1 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{2}R_2]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The special solution $(1,0,1)'$ is a basis of $\mathcal{N}(A - 2I)$.

Both eigenspaces are lines, although p_A has the factor $(t-2)^2$.

Theorem 8. (a) (**Fundamental theorem of algebra**; proved in complex analysis, not in this course.) Every polynomial of degree at least 1 has a root in $\mathbb{C}$.

(b) Every monic polynomial f of degree $n \geqslant 1$ is a product

$$f = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n) \qquad (\lambda_1, \ldots, \lambda_n \in \mathbb{C})$$

and the list $\lambda_1, \ldots, \lambda_n$ is unique up to order: if also $f = (t - \mu_1) \cdots (t - \mu_n)$, then $\mu_1, \ldots, \mu_n$ are $\lambda_1, \ldots, \lambda_n$ in some order.

(c) In (b), the roots of f are exactly $\lambda_1, \ldots, \lambda_n$.

Proof. (b)

- Existence, by induction on n. For $n = 1$, $f = t + c_0 = t - (-c_0)$. Let $n \geqslant 2$. By (a), f has a root λ_n, so $f = (t - \lambda_n)g$ (Proposition 4(c)); here $g \neq 0$, as $f \neq 0$, so g is monic of degree $n - 1$ (Proposition 4(a)). By the induction hypothesis, $g = (t - \lambda_1) \cdots (t - \lambda_{n-1})$.
- Uniqueness, by induction on n. For $n = 1$, $t - \lambda_1 = t - \mu_1$ gives $\lambda_1 = \mu_1$ (constant terms). Let $n \geqslant 2$. The value of both products at λ_n (Proposition 4(b)) gives $0 = (\lambda_n - \mu_1) \cdots (\lambda_n - \mu_n)$, so $\lambda_n = \mu_k$ for some k (a product is 0 only if a factor is 0).
- Renumber the μ_i so that $k = n$, and cancel the factor $t - \lambda_n \neq 0$ (Proposition 4(a)): $(t - \lambda_1) \cdots (t - \lambda_{n-1}) = (t - \mu_1) \cdots (t - \mu_{n-1})$. By the induction hypothesis, $\mu_1, \ldots, \mu_{n-1}$ are $\lambda_1, \ldots, \lambda_{n-1}$ in some order, and $\mu_n = \lambda_n$.

(c) By Proposition 4(b), $f(z) = (z - \lambda_1) \cdots (z - \lambda_n)$, which is 0 if and only if $z = \lambda_i$ for some i. □

Definition 9. Let $A \in \mathbb{R}^{n \times n}$. By Theorem 6(b) and Theorem 8(b),

$$p_A(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n)$$

for complex numbers $\lambda_1, \ldots, \lambda_n$, unique up to order. These n numbers are the **eigenvalues** of A, listed **with multiplicity**; by Theorem 8(c) they are the roots of p_A. For $c \in \mathbb{C}$, the **algebraic multiplicity** $a_A(c)$ is the number of indices i with $\lambda_i = c$ ($a_A(c) = 0$ if c is not an eigenvalue); so if $c_1, \ldots, c_k$ are the distinct eigenvalues, then $p_A = (t - c_1)^{a_A(c_1)} \cdots (t - c_k)^{a_A(c_k)}$ and $a_A(c_1) + \cdots + a_A(c_k) = n$. An eigenvalue c is **simple** if $a_A(c) = 1$ and **multiple** if $a_A(c) \geqslant 2$. A real number is an eigenvalue in this sense if and only if it is one in the sense of Definition 1, since both mean that it is a real root of p_A (Theorem 6(c)); these are the **real eigenvalues**. Eigenvectors and eigenspaces belong to real eigenvalues only. From now on "eigenvalue" has the meaning of this definition; Theorem 2 and Theorem 6(c) describe the real eigenvalues.

In Example 7 the eigenvalues, listed with multiplicity, are 0, 2, 2: $a_A(0) = 1$ and $a_A(2) = 2$.

Example 10. Find the eigenvalues of $A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$ and the eigenspace of each real eigenvalue.

Solution.

- For $t \in \mathbb{R}$, column 3 of $tI - A$ has the entry $t-1$ in position $(3,3)$ and 0 elsewhere; expanding along it (Fact F8),

$$\det(tI - A) = \det \begin{pmatrix} t-2 & 1 & 0 \\ -1 & t-2 & 0 \\ -1 & -1 & t-1 \end{pmatrix} = (t-1)\det \begin{pmatrix} t-2 & 1 \\ -1 & t-2 \end{pmatrix} = (t-1)\Big((t-2)^2+1\Big)$$

- Since $(t-2-\mathrm{i})(t-2+\mathrm{i}) = (t-2)^2 - \mathrm{i}^2 = (t-2)^2+1$, $p_A(t) = (t-1)\Big(t-(2+\mathrm{i})\Big)\Big(t-(2-\mathrm{i})\Big)$: the eigenvalues are 1, $2+\mathrm{i}$ and $2-\mathrm{i}$, each simple (Definition 9); $2+\mathrm{i}$ and $2-\mathrm{i}$ are conjugate. The only real eigenvalue is 1.
- Row-reduce $A - I$:

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  So $x_1 = x_2 = 0$ and x_3 is free: $\mathcal{N}(A-I) = \operatorname{span}\{e_3\}$. The eigenvalues $2 \pm \mathrm{i}$ have no eigenvectors in $\mathbb{R}^3$ (Definition 9).

So a real matrix may have eigenvalues that are not real numbers.

4. Similar and triangular matrices

Definition 11. A matrix $B \in \mathbb{R}^{n\times n}$ is **similar** to $A \in \mathbb{R}^{n\times n}$ if $B = P^{-1}AP$ for some invertible $P \in \mathbb{R}^{n\times n}$. Then $A = PBP^{-1} = (P^{-1})^{-1}BP^{-1}$, so A is similar to B.

Theorem 12. Let $A, P \in \mathbb{R}^{n\times n}$ with P invertible, and let $B = P^{-1}AP$.

(a) $p_B = p_A$. Hence A and B have the same eigenvalues with the same algebraic multiplicities, $\operatorname{tr} B = \operatorname{tr} A$ and $\det B = \det A$.

(b) For every $c \in \mathbb{R}$, $x \mapsto P^{-1}x$ maps $\mathcal{N}(A-cI)$ onto $\mathcal{N}(B-cI)$, with inverse $y \mapsto Py$; so x is an eigenvector of A belonging to c if and only if $P^{-1}x$ is an eigenvector of B belonging to c. Moreover $\dim \mathcal{N}(B-cI) = \dim \mathcal{N}(A-cI)$.

Proof. (a)

- For $c \in \mathbb{R}$, $cI - B = P^{-1}(cI)P - P^{-1}AP = P^{-1}(cI-A)P$ (Fact F1), so $\det(cI-B) = \det P^{-1} \cdot \det(cI-A) \cdot \det P = \det(cI-A)$, since $\det P^{-1} \det P = \det(P^{-1}P) = \det I = 1$ (Fact F7).
- So $p_B(c) = p_A(c)$ for every real c, and $p_B = p_A$ (Theorem 6(a)).
- The eigenvalues of A and B come from the same factorization (Definition 9); $-\operatorname{tr} B$ and $-\operatorname{tr} A$ are the same coefficient of t^{n-1}, and $(-1)^n \det B$, $(-1)^n \det A$ the same constant term (Theorem 6(b)).

(b)

- $B - cI = P^{-1}(A - cI)P$ and $A - cI = P(B - cI)P^{-1}$. If $(A - cI)x = 0$, then $(B - cI)P^{-1}x = P^{-1}(A - cI)x = 0$; if $(B - cI)y = 0$, then $(A - cI)Py = P(B - cI)y = 0$.
- The two maps are inverse to each other ($PP^{-1}x = x$, $P^{-1}Py = y$), and $P^{-1}x \neq 0$ if and only if $x \neq 0$.
- $\operatorname{rank}(B - cI) = \operatorname{rank}\Big(P^{-1}(A - cI)P\Big) = \operatorname{rank}(A - cI)$, so the two null spaces have the same dimension (Fact F6). □

Proposition 13. (a) Let $A = \begin{pmatrix} B & K \\ 0 & D \end{pmatrix}$ or $A = \begin{pmatrix} B & 0 \\ K & D \end{pmatrix}$ with $B \in \mathbb{R}^{p\times p}$ and $D \in \mathbb{R}^{q\times q}$. Then $p_A = p_B p_D$: the eigenvalues of A, listed with multiplicity, are those of B together with those of D.

(b) If $A \in \mathbb{R}^{n\times n}$ is upper or lower triangular, then $p_A = (t - a_{11}) \cdots (t - a_{nn})$: the eigenvalues of A, listed with multiplicity, are $a_{11}, \ldots, a_{nn}$.

Proof. (a)

- For $c \in \mathbb{R}$, $cI - A$ is block triangular of the same kind, with diagonal blocks $cI_p - B$ and $cI_q - D$ (Fact F1). By Fact F9, $\det(cI - A) = \det(cI_p - B)\det(cI_q - D) = p_B(c)p_D(c)$, the value of $p_B p_D$ at c (Proposition 4(b)). So $p_A = p_B p_D$ (Theorem 6(a)).
- If $p_B = (t - \beta_1) \cdots (t - \beta_p)$ and $p_D = (t - \delta_1) \cdots (t - \delta_q)$ (Definition 9), then p_A is the product of these $p + q$ factors $t - c$, and by Theorem 8(b) the eigenvalues of A are $\beta_1, \ldots, \beta_p, \delta_1, \ldots, \delta_q$.

(b)

- For $c \in \mathbb{R}$, $cI - A$ is triangular of the same kind with diagonal entries $c - a_{ii}$, so $\det(cI - A) = (c - a_{11}) \cdots (c - a_{nn})$ (Fact F7), the value of $(t - a_{11}) \cdots (t - a_{nn})$ at c.
- As in (a), $p_A = (t - a_{11}) \cdots (t - a_{nn})$, and Definition 9 gives the eigenvalues. □

5. Eigenvalues of related matrices; trace and determinant

Proposition 14. Let $A \in \mathbb{R}^{n\times n}$.

(a) If $Ax = \lambda x$ with $\lambda \in \mathbb{R}$ and $x \neq 0$, then $A^k x = \lambda^k x$ $(k = 0, 1, 2, \ldots)$ and $f(A)x = f(\lambda)x$ for every real polynomial f; in particular $(cA)x = (c\lambda)x$ and $(A + cI)x = (\lambda + c)x$ $(c \in \mathbb{R})$. So x is an eigenvector of A^k, $f(A)$, cA and $A + cI$, belonging to λ^k, $f(\lambda)$, $c\lambda$ and $\lambda + c$.

(b) If, moreover, A is invertible, then $\lambda \neq 0$ and $A^{-1}x = \lambda^{-1}x$.

(c) If A has the eigenvalues $\lambda_1, \ldots, \lambda_n$ (with multiplicity) and $c \in \mathbb{R}$, then $A + cI$ has the eigenvalues $\lambda_1 + c, \ldots, \lambda_n + c$, and, if $c \neq 0$, cA has the eigenvalues $c\lambda_1, \ldots, c\lambda_n$.

(d) $p_{A'} = p_A$: A' and A have the same eigenvalues with the same algebraic multiplicities.

Proof. (a)

- By induction on k: $A^0 x = x = \lambda^0 x$, and if $A^k x = \lambda^k x$, then $A^{k+1}x = A^k(Ax) = A^k(\lambda x) = \lambda A^k x = \lambda^{k+1}x$ (Fact F1).
- If $f = \sum_k c_k t^k$, then $f(A)x = \sum_k c_k A^k x = \sum_k c_k \lambda^k x = f(\lambda)x$. For $f = ct$ and $f = t + c$ this gives $(cA)x = c\lambda x$ and $(A + cI)x = (\lambda + c)x$. In each case $x \neq 0$.

(b) If $\lambda = 0$, then $Ax = 0$ with $x \neq 0$, and A would not be invertible (Fact F4) ✘ So $\lambda \neq 0$, and $x = A^{-1}(Ax) = A^{-1}(\lambda x) = \lambda A^{-1}x$, that is, $A^{-1}x = \lambda^{-1}x$.

(c)

- For $s \in \mathbb{R}$, $\det\big(sI - (A+cI)\big) = \det\big((s-c)I - A\big) = p_A(s-c) = (s-c-\lambda_1)\cdots(s-c-\lambda_n)$ (Theorem 6(a), Proposition 4(b)), the value at s of $\big(t-(\lambda_1+c)\big)\cdots\big(t-(\lambda_n+c)\big)$. By Theorem 6(a) this product is p_{A+cI}, and Definition 9 gives the eigenvalues.
- Let $c \neq 0$. For $s \in \mathbb{R}$, $\det(sI - cA) = \det\big(c(\frac{s}{c}I - A)\big) = c^n p_A(\frac{s}{c}) = c^n(\frac{s}{c} - \lambda_1)\cdots(\frac{s}{c} - \lambda_n) = (s - c\lambda_1)\cdots(s - c\lambda_n)$, as $\det(cB) = c^n \det B$ (Fact F7); conclude as before.

(d) For $c \in \mathbb{R}$, $(cI - A)' = cI - A'$ (Fact F3), so $\det(cI - A') = \det(cI - A)$, as $\det B' = \det B$ (Fact F7). By Theorem 6(a), $p_{A'} = p_A$. □

Theorem 15. Let $A \in \mathbb{R}^{n\times n}$ have the eigenvalues $\lambda_1, \ldots, \lambda_n$, listed with multiplicity. Then

$$\operatorname{tr} A = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \qquad \det A = \lambda_1\lambda_2\cdots\lambda_n$$

Hence A is invertible if and only if 0 is not an eigenvalue of A.

Proof.
- $p_A = (t-\lambda_1)\cdots(t-\lambda_n)$ (Definition 9). By Proposition 4(e), its coefficient of t^{n-1} is $-(\lambda_1 + \cdots + \lambda_n)$ and its constant term is $(-1)^n\lambda_1\cdots\lambda_n$.
- By Theorem 6(b), the same two coefficients are $-\operatorname{tr} A$ and $(-1)^n \det A$.
- A is invertible if and only if $\det A \neq 0$ (Fact F7), that is, if and only if no λ_i is 0. □

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & 3 \end{smallmatrix}\right)$ of Example 7, with the eigenvalues 0, 2, 2: $\operatorname{tr} A = 1 + 0 + 3 = 4 = 0 + 2 + 2$ and $\det A = 0 = 0 \cdot 2 \cdot 2$, so A is not invertible. For $A = \left(\begin{smallmatrix} 2 & -1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ of Example 10, with the eigenvalues 1, $2+\mathrm{i}$, $2-\mathrm{i}$: $\operatorname{tr} A = 2 + 2 + 1 = 5 = 1 + (2+\mathrm{i}) + (2-\mathrm{i})$ and $\det A = 5 = 1 \cdot (2+\mathrm{i})(2-\mathrm{i})$, as $(2+\mathrm{i})(2-\mathrm{i}) = 4 - \mathrm{i}^2 = 5$.

6. Eigenvectors of distinct eigenvalues; multiplicities

Theorem 16. Let $v_1, \ldots, v_m$ be eigenvectors of $A \in \mathbb{R}^{n\times n}$ belonging to real eigenvalues $\lambda_1, \ldots, \lambda_m$ that are pairwise distinct ($\lambda_i \neq \lambda_j$ for $i \neq j$). Then $v_1, \ldots, v_m$ are linearly independent. In particular, if A has n distinct real eigenvalues, then eigenvectors belonging to them form a basis of $\mathbb{R}^n$.

Proof. Induction on m. For $m = 1$, $c_1v_1 = 0$ with $v_1 \neq 0$ gives $c_1 = 0$ (Fact F5). Let $m \geqslant 2$, assume the statement for $m-1$ eigenvectors, and let $c_1v_1 + \cdots + c_mv_m = 0$.

- Multiplying by λ_m gives $\sum_i c_i\lambda_m v_i = 0$; multiplying by A gives $\sum_i c_i\lambda_i v_i = 0$, since $Av_i = \lambda_i v_i$ (Fact F1).
- Subtracting, the v_m term vanishes: $c_1(\lambda_m - \lambda_1)v_1 + \cdots + c_{m-1}(\lambda_m - \lambda_{m-1})v_{m-1} = 0$.
- By the induction hypothesis, $c_i(\lambda_m - \lambda_i) = 0$ for $i < m$, and $c_i = 0$ since $\lambda_m \neq \lambda_i$.
- Then $c_m v_m = 0$ with $v_m \neq 0$, so $c_m = 0$ (Fact F5).

If A has n distinct real eigenvalues, eigenvectors belonging to them are n independent vectors of $\mathbb{R}^n$, which has dimension n; so they form a basis (Fact F5). □

Definition 17. Let $A \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}$. The **geometric multiplicity** of c is

$$g_A(c) = \dim \mathcal{N}(A - cI) = n - \operatorname{rank}(A - cI)$$

(Fact F6). So $g_A(c) \geqslant 1$ if c is a real eigenvalue of A, and $g_A(c) = 0$ otherwise (Theorem 2, (i)⇔(iv)).

By Theorem 12(a),(b), similar matrices have the same algebraic and geometric multiplicities. In Example 7, $g_A(0) = a_A(0) = 1$, but $g_A(2) = 1 < 2 = a_A(2)$.

Theorem 18. If c is a real eigenvalue of $A \in \mathbb{R}^{n\times n}$, then $1 \leqslant g_A(c) \leqslant a_A(c)$. In particular, if c is simple, then $g_A(c) = 1$, that is, $\operatorname{rank}(A - cI) = n - 1$.

Proof.
- $g = g_A(c) \geqslant 1$ (Definition 17). Extend a basis $x_1, \dots, x_g$ of $\mathcal{N}(A - cI)$ to a basis $x_1, \dots, x_n$ of $\mathbb{R}^n$ (Fact F5). Then $P = [x_1 \ \cdots \ x_n]$ has independent columns, so it is invertible (Fact F4).
- For $j \leqslant g$, $Ax_j = cx_j$; put $m_{ij} = c\delta_{ij}$. For $j > g$, write $Ax_j = \sum_{i=1}^n m_{ij}x_i$, as $x_1, \dots, x_n$ span $\mathbb{R}^n$. With $M = (m_{ij})$, $\operatorname{col}_j(AP) = Ax_j = \sum_i m_{ij}x_i = P\operatorname{col}_j(M)$ for every j (Fact F2); so $AP = PM$ and $M = P^{-1}AP$.
- The first g columns of M are $ce_1, \dots, ce_g$: $M = cI_n$ if $g = n$, and $M = \left(\begin{smallmatrix} cI_g & K \\ 0 & D \end{smallmatrix}\right)$ with $D \in \mathbb{R}^{(n-g)\times(n-g)}$ if $g < n$. By Theorem 12(a) and Proposition 13(a),(b), the eigenvalues of A are those of M: c (g times), together with those of D if $g < n$. Hence $a_A(c) \geqslant g$.
- If c is simple, then $1 \leqslant g_A(c) \leqslant a_A(c) = 1$, and $\operatorname{rank}(A - cI) = n - g_A(c) = n - 1$. □

7. Diagonalization

Definition 19. A matrix $A \in \mathbb{R}^{n\times n}$ is **diagonalizable** if $P^{-1}AP$ is diagonal for some invertible $P \in \mathbb{R}^{n\times n}$, that is, if A is similar to a diagonal matrix; P is then said to **diagonalize** A. Here $\operatorname{diag}(\lambda_1, \dots, \lambda_n)$ is the diagonal matrix with the diagonal entries $\lambda_1, \dots, \lambda_n$. Let $P = [u_1 \ \cdots \ u_n] \in \mathbb{R}^{n\times n}$ be invertible and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_j \in \mathbb{R}$. Then

$$P^{-1}AP = \Lambda \iff AP = P\Lambda \iff A = P\Lambda P^{-1} \iff Au_j = \lambda_j u_j \quad (1 \leqslant j \leqslant n)$$

Consequently A is diagonalizable if and only if $\mathbb{R}^n$ has a basis consisting of eigenvectors of A. If $P^{-1}AP = \Lambda$, then u_j is an eigenvector belonging to λ_j, and $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A, listed with multiplicity; in particular all eigenvalues of a diagonalizable matrix are real.

Proof.
- If $P^{-1}AP = \Lambda$, then $AP = P(P^{-1}AP) = P\Lambda$ and $A = P\Lambda P^{-1}$; if $AP = P\Lambda$ or $A = P\Lambda P^{-1}$, then $P^{-1}AP = \Lambda$ (Fact F1).
- $\operatorname{col}_j(AP) = Au_j$ and $\operatorname{col}_j(P\Lambda) = P(\lambda_j e_j) = \lambda_j u_j$ (Fact F2); two matrices are equal if and only if their columns are.
- If $P^{-1}AP = \Lambda$, the columns of P form a basis of $\mathbb{R}^n$ (Fact F4), so they are nonzero (Fact F5) and, as $Au_j = \lambda_j u_j$, eigenvectors. Conversely, if eigenvectors $u_1, \dots, u_n$ with $Au_j = \lambda_j u_j$ form a basis of $\mathbb{R}^n$, then $P = [u_1 \ \cdots \ u_n]$ is invertible (Fact F4) and $P^{-1}AP = \Lambda$.
- By Theorem 12(a) and Proposition 13(b), $p_A = p_\Lambda = (t - \lambda_1)\cdots(t - \lambda_n)$; so $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (Definition 9). □

Lemma 20. Let $c_1, \dots, c_k$ ($k \geqslant 1$) be distinct real eigenvalues of $A \in \mathbb{R}^{n\times n}$, and $W_i = \mathcal{N}(A - c_i I)$.

(a) If $w_i \in W_i$ ($1 \leqslant i \leqslant k$) and $w_1 + \cdots + w_k = 0$, then $w_i = 0$ for every i.

(b) If $v_{i1}, \dots, v_{id_i}$ is a basis of W_i for each i, then the $d_1 + \cdots + d_k$ vectors $v_{11}, \dots, v_{1d_1}, \dots, v_{k1}, \dots, v_{kd_k}$ are linearly independent.

Proof. (a)
- For a real polynomial f and $w \in W_i$, $f(A)w = f(c_i)w$ (Proposition 14(a) if $w \neq 0$; both sides are 0 if $w = 0$). Applying $f(A)$ to $w_1 + \cdots + w_k = 0$ gives $f(c_1)w_1 + \cdots + f(c_k)w_k = 0$ (Fact F1).
- For a fixed i, let $f_i = \prod_{j\neq i} \frac{t - c_j}{c_i - c_j}$, the product over $j \in \{1, \dots, k\}$, $j \neq i$ ($f_1 = 1$ if $k = 1$). By Proposition 4(b), $f_i(c_l) = \prod_{j\neq i} \frac{c_l - c_j}{c_i - c_j}$, which is 1 for $l = i$ and 0 for $l \neq i$ (the factor $j = l$ vanishes).

- With $f = f_i$, the relation of the first item reads $w_i = 0$.

(b) A linear relation among these vectors has the form $w_1 + \cdots + w_k = 0$ with $w_i = \sum_l a_{il} v_{il} \in W_i$. By (a), every $w_i = 0$, and then every $a_{il} = 0$, since $v_{i1}, \ldots, v_{id_i}$ are independent. □

Theorem 21. Let $A \in \mathbb{R}^{n\times n}$, let $c_1, \ldots, c_k$ be the distinct real eigenvalues of A ($k \geqslant 0$; an empty sum is 0, an empty product 1), and let $W_i = \mathcal{N}(A - c_i I)$. The following are equivalent:

(i) A is diagonalizable;

(ii) $p_A = (t - c_1)^{d_1} \cdots (t - c_k)^{d_k}$ for some positive integers $d_1, \ldots, d_k$, and $\dim W_i = d_i$ for each i;

(iii) $\dim W_1 + \cdots + \dim W_k = n$.

In other words, A is diagonalizable if and only if all its eigenvalues are real and $g_A(c) = a_A(c)$ for each of them, that is, $\operatorname{rank}(A - c_i I) = n - a_A(c_i)$ for each i. In particular, an $n \times n$ matrix with n distinct real eigenvalues is diagonalizable.

Proof.
- (i)⇒(ii): let $P^{-1}AP = \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By Definition 19, $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A; they are real, so each is one of $c_1, \ldots, c_k$. Let d_i be the number of j with $\lambda_j = c_i$; $d_i \geqslant 1$, as c_i is an eigenvalue. Then $p_A = (t - \lambda_1) \cdots (t - \lambda_n) = (t - c_1)^{d_1} \cdots (t - c_k)^{d_k}$.
- Still (i)⇒(ii): $\dim W_i = \dim \mathcal{N}(\Lambda - c_i I)$ (Theorem 12(b)). The jth entry of $(\Lambda - c_i I)x$ is $(\lambda_j - c_i)x_j$, so $\mathcal{N}(\Lambda - c_i I)$ consists of the x with $x_j = 0$ whenever $\lambda_j \neq c_i$. These are the linear combinations of the d_i vectors e_j with $\lambda_j = c_i$, which are linearly independent (a combination of them is 0 only if every coefficient, an entry of it, is 0); so they form a basis, and $\dim W_i = d_i$.
- (ii)⇒(iii): $\dim W_1 + \cdots + \dim W_k = d_1 + \cdots + d_k = \deg p_A = n$ (Proposition 4(a), Theorem 6(b)).
- (iii)⇒(i): as $n \geqslant 1$, $k \geqslant 1$. By Lemma 20(b), bases of $W_1, \ldots, W_k$ together give n independent eigenvectors of A; they form a basis of $\mathbb{R}^n$ (Fact F5), and A is diagonalizable (Definition 19).
- In (ii), $d_i = a_A(c_i)$, as the list of eigenvalues in Definition 9 is unique up to order (Theorem 8(b)), and p_A has no other roots (Theorem 8(c)); so (ii) says that all eigenvalues are real and $g_A(c_i) = a_A(c_i)$. Conversely, if all eigenvalues are real, then $p_A = (t - c_1)^{a_A(c_1)} \cdots (t - c_k)^{a_A(c_k)}$ (Definition 9), and (ii) holds when $g_A(c_i) = a_A(c_i)$ for each i. By Definition 17, $g_A(c_i) = a_A(c_i)$ means $\operatorname{rank}(A - c_i I) = n - a_A(c_i)$.
- If A has n distinct real eigenvalues, eigenvectors belonging to them form a basis of $\mathbb{R}^n$ (Theorem 16), so A is diagonalizable (Definition 19). □

For example, the matrix of Example 7 is not diagonalizable: its eigenvalues 0, 2, 2 are real, but $g_A(2) = 1 < 2 = a_A(2)$. Nor is the matrix of Example 10, which has non-real eigenvalues.

Example 22. Show that $A = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix}$ is diagonalizable, and find an invertible P and a diagonal Λ with $P^{-1}AP = \Lambda$.

Solution.

- For $t \in \mathbb{R}$, subtract column 3 from column 2, and take the factor $t - 2$ out of column 2 (Fact F7):

$$\det(tI - A) = \det \begin{pmatrix} t-5 & 6 & 6 \\ 1 & t-4 & -2 \\ -3 & 6 & t+4 \end{pmatrix} = \det \begin{pmatrix} t-5 & 0 & 6 \\ 1 & t-2 & -2 \\ -3 & 2-t & t+4 \end{pmatrix}$$
$$= (t-2) \det \begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -3 & -1 & t+4 \end{pmatrix}$$

Add row 2 to row 3 (Fact F7) and expand along column 2, whose only nonzero entry is the $(2,2)$ entry 1 (Fact F8):

$$\det(tI-A) = (t-2)\det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -2 & 0 & t+2 \end{pmatrix} = (t-2)\det\begin{pmatrix} t-5 & 6 \\ -2 & t+2 \end{pmatrix} = (t-2)(t^2-3t+2)$$

As $t^2-3t+2=(t-1)(t-2)$, $p_A=(t-1)(t-2)^2$: the eigenvalues are 1, 2, 2, with $a_A(1)=1$ and $a_A(2)=2$.

- $\lambda=1$:

$$A-I = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow{R_1\leftrightarrow R_2} \begin{pmatrix} -1 & 3 & 2 \\ 4 & -6 & -6 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow[R_3\leftarrow R_3+3R_1]{R_2\leftarrow R_2+4R_1} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 6 & 2 \\ 0 & 3 & 1 \end{pmatrix}$$
$$\xrightarrow{R_2\leftrightarrow R_3} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 6 & 2 \end{pmatrix} \xrightarrow{R_3\leftarrow R_3-2R_2} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank}(A-I)=2$ (Fact F6) and $g_A(1)=1$ (Definition 17). Row operations do not change the null space (Fact F6), so $\mathcal{N}(A-I)$ is given by the equations $-x_1+3x_2+2x_3=0$, $3x_2+x_3=0$; with $x_3=3$ they give $x_2=-1$ and $x_1=3$: the nonzero vector $\alpha_1=(3,-1,3)'$ of the line $\mathcal{N}(A-I)$ is a basis of it (Fact F5).

- $\lambda=2$:

$$A-2I = \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow{R_1\leftrightarrow R_2} \begin{pmatrix} -1 & 2 & 2 \\ 3 & -6 & -6 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow[R_3\leftarrow R_3+3R_1]{R_2\leftarrow R_2+3R_1} \begin{pmatrix} -1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_1\leftarrow -R_1} \begin{pmatrix} 1 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So $g_A(2)=3-1=2$, and the special solutions $\alpha_2=(2,1,0)'$, $\alpha_3=(2,0,1)'$ form a basis of $\mathcal{N}(A-2I)$ (Fact F6).

- $g_A(1)+g_A(2)=3$, so A is diagonalizable (Theorem 21), and $\alpha_1,\alpha_2,\alpha_3$ form a basis of $\mathbb{R}^3$ (Lemma 20(b), Fact F5). With

$$P=[\alpha_1\ \alpha_2\ \alpha_3]=\begin{pmatrix} 3 & 2 & 2 \\ -1 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix}, \qquad \Lambda=\operatorname{diag}(1,2,2), \qquad AP=\begin{pmatrix} 3 & 4 & 4 \\ -1 & 2 & 0 \\ 3 & 0 & 2 \end{pmatrix}=P\Lambda$$

we have $P^{-1}AP=\Lambda$ (Definition 19).

Proposition 23. Let $A=P\Lambda P^{-1}$ with $P\in\mathbb{R}^{n\times n}$ invertible and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_n)$, $\lambda_i\in\mathbb{R}$.

(a) $A^k=P\Lambda^kP^{-1}=P\operatorname{diag}(\lambda_1^k,\dots,\lambda_n^k)P^{-1}$ for $k=0,1,2,\dots$

(b) $f(A)=Pf(\Lambda)P^{-1}=P\operatorname{diag}\big(f(\lambda_1),\dots,f(\lambda_n)\big)P^{-1}$ for every real polynomial f.

(c) A is invertible if and only if every $\lambda_i\neq 0$. Then $A^{-1}=P\Lambda^{-1}P^{-1}$, and for $k\geqslant 1$ the power $A^{-k}=(A^{-1})^k$ equals $P\operatorname{diag}(\lambda_1^{-k},\dots,\lambda_n^{-k})P^{-1}$.

Proof. (a)

- By induction on k: $A^0=I=P\Lambda^0P^{-1}$, and if $A^k=P\Lambda^kP^{-1}$, then $A^{k+1}=A^kA=P\Lambda^kP^{-1}P\Lambda P^{-1}=P\Lambda^{k+1}P^{-1}$ (Fact F1).

- For diagonal matrices, $\operatorname{diag}(a_1,\dots,a_n)\operatorname{diag}(b_1,\dots,b_n) = \operatorname{diag}(a_1b_1,\dots,a_nb_n)$: column j of the product is $\operatorname{diag}(a_1,\dots,a_n)(b_je_j) = a_jb_je_j$ (Fact F2). By induction on k, $\Lambda^k = \operatorname{diag}(\lambda_1^k,\dots,\lambda_n^k)$.

(b) If $f = \sum_k c_k t^k$, then by (a) and Fact F1, $f(A) = \sum_k c_k P\Lambda^k P^{-1} = Pf(\Lambda)P^{-1}$; and $f(\Lambda) = \sum_k c_k \operatorname{diag}(\lambda_1^k,\dots,\lambda_n^k) = \operatorname{diag}\Big(f(\lambda_1),\dots,f(\lambda_n)\Big)$, entry by entry.

(c)

- The eigenvalues of A are $\lambda_1,\dots,\lambda_n$ (Definition 19), so A is invertible if and only if every $\lambda_i \neq 0$ (Theorem 15).
- Then $\Lambda^{-1} = \operatorname{diag}(\lambda_1^{-1},\dots,\lambda_n^{-1})$, since the product of the two diagonal matrices is I by the rule in (a); and $A^{-1} = (P^{-1})^{-1}\Lambda^{-1}P^{-1} = P\Lambda^{-1}P^{-1}$ (Fact F4). Applying (a) to $A^{-1} = P\Lambda^{-1}P^{-1}$ gives $A^{-k} = P(\Lambda^{-1})^kP^{-1} = P\operatorname{diag}(\lambda_1^{-k},\dots,\lambda_n^{-k})P^{-1}$. □

For $A = \left(\begin{smallmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{smallmatrix}\right) = P\operatorname{diag}(1,2,2)P^{-1}$ of Example 22, put $E = P\operatorname{diag}(1,0,0)P^{-1}$.

- $\operatorname{diag}(1,2,2) = 2I - \operatorname{diag}(1,0,0)$, so $A = P(2I - \operatorname{diag}(1,0,0))P^{-1} = 2I - E$ (Fact F1), and $E = 2I - A$.
- For $k \geqslant 0$, $\operatorname{diag}(1,2^k,2^k) = \operatorname{diag}(1,0,0) + 2^k\Big(I - \operatorname{diag}(1,0,0)\Big)$, so by Proposition 23(a)

$$A^k = P\operatorname{diag}(1,2^k,2^k)P^{-1} = E + 2^k(I-E) = (2-2^k)I + (2^k-1)A$$

- All eigenvalues are nonzero, so A is invertible, and by Proposition 23(c) the same computation holds for every negative integer k. For $k=-1$:

$$A^{-1} = \tfrac{3}{2}I - \tfrac{1}{2}A = \tfrac{1}{2}(3I-A) = \tfrac{1}{2}\begin{pmatrix} -2 & 6 & 6 \\ 1 & -1 & -2 \\ -3 & 6 & 7 \end{pmatrix}$$

8. The products AB and BA

Use: $X'X$ and XX' for a data matrix X (singular value decomposition and principal components, later).

Theorem 24. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$.

(a) $t^n p_{AB}(t) = t^m p_{BA}(t)$.

(b) If $m \leqslant n$, then $p_{BA} = t^{n-m}p_{AB}$: the eigenvalues of BA, listed with multiplicity, are those of AB together with $n-m$ zeros. So AB and BA have the same nonzero eigenvalues, with the same algebraic multiplicities.

(c) Let $\lambda \neq 0$ be real. Then $x \mapsto Bx$ maps $\mathcal{N}(AB-\lambda I_m)$ into $\mathcal{N}(BA-\lambda I_n)$, and $Bx \neq 0$ for every $x \neq 0$ in $\mathcal{N}(AB-\lambda I_m)$; moreover $g_{AB}(\lambda) = g_{BA}(\lambda)$.

Proof. (a)

- For $t \in \mathbb{R}$ let $M = \begin{pmatrix} tI_m & A \\ B & I_n \end{pmatrix}$. By Fact F9, with the invertible block I_n, $\det M = \det I_n \det(tI_m - AI_n^{-1}B) = \det(tI_m - AB) = p_{AB}(t)$.
- For $t \neq 0$ the block tI_m is invertible, with inverse $t^{-1}I_m$, and Fact F9 gives $\det M = \det(tI_m)\det(I_n - t^{-1}BA)$. Here $\det(tI_m) = t^m \det I_m = t^m$ and $\det(I_n - t^{-1}BA) = \det\Big(t^{-1}(tI_n - BA)\Big) = t^{-n}p_{BA}(t)$, as $\det(cG) = c^k \det G$ for $G \in \mathbb{R}^{k\times k}$ (Fact F7).

- So $t^n p_{AB}(t) = t^m p_{BA}(t)$ for every real $t \neq 0$. By Proposition 4(b),(d), the polynomials $t^n p_{AB}$ and $t^m p_{BA}$ are equal.

(b) By (a), $t^m\big(t^{n-m}p_{AB}\big) = t^m p_{BA}$, so $p_{BA} = t^{n-m}p_{AB}$ (Proposition 4(a)). If $p_{AB} = (t-\mu_1)\cdots(t-\mu_m)$ (Definition 9), then $p_{BA} = (t-0)^{n-m}(t-\mu_1)\cdots(t-\mu_m)$, a product of n factors $t-c$. By Theorem 8(b), the eigenvalues of BA are $\mu_1, \ldots, \mu_m$ and $n-m$ zeros; so $a_{BA}(c) = a_{AB}(c)$ for every $c \neq 0$.

(c)

- If $ABx = \lambda x$, then $BA(Bx) = B(ABx) = \lambda Bx$, so $Bx \in \mathcal{N}(BA - \lambda I_n)$; and if $Bx = 0$, then $\lambda x = ABx = 0$, so $x = 0$ (Fact F5).
- Let $x_1, \ldots, x_g$ be a basis of $\mathcal{N}(AB - \lambda I_m)$, $g = g_{AB}(\lambda)$. If $\sum_i c_i Bx_i = 0$, then $B(\sum_i c_i x_i) = 0$ with $\sum_i c_i x_i \in \mathcal{N}(AB - \lambda I_m)$, so $\sum_i c_i x_i = 0$ by the first item, and every $c_i = 0$. Thus $\mathcal{N}(BA - \lambda I_n)$ contains g linearly independent vectors, and $g_{BA}(\lambda) \geqslant g_{AB}(\lambda)$ (Fact F5).
- The same argument with A and B exchanged gives $g_{AB}(\lambda) \geqslant g_{BA}(\lambda)$. □

Example 25. For $n \geqslant 2$, take $A = \mathbf{1}_n' \in \mathbb{R}^{1\times n}$ and $B = \mathbf{1}_n \in \mathbb{R}^{n\times 1}$ in Theorem 24 ($m = 1$): $AB = \mathbf{1}_n'\mathbf{1}_n = (n)$, a 1×1 matrix, and $BA = \mathbf{1}_n\mathbf{1}_n' = J_n$ (Fact F3). As $p_{(n)} = t - n$ (Theorem 6(b)), Theorem 24(b) gives $p_{J_n} = t^{n-1}(t-n)$: the eigenvalues of J_n are n (simple) and 0 ($n-1$ times). Indeed $J_n\mathbf{1}_n = \mathbf{1}_n(\mathbf{1}_n'\mathbf{1}_n) = n\mathbf{1}_n$, and $J_n x = \mathbf{1}_n(\mathbf{1}_n' x) = 0$ for every centered x, that is, $\mathbf{1}_n' x = 0$.

Example 26. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \qquad AA' = \begin{pmatrix} 2 & 1 & 1 & -1 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix}$$

Find the eigenvalues of $A'A$ and of AA', and eigenvectors for the nonzero eigenvalues of AA'.

Solution.

- For $t \in \mathbb{R}$, expanding along the first row (Fact F8),

$$\det(tI_3 - A'A) = \det\begin{pmatrix} t-3 & -1 & 0 \\ -1 & t-2 & -1 \\ 0 & -1 & t-3 \end{pmatrix} = (t-3)\Big((t-2)(t-3)-1\Big) - (t-3)$$
$$= (t-3)(t^2 - 5t + 4) = (t-1)(t-3)(t-4)$$

  so $p_{A'A}(t) = (t-1)(t-3)(t-4)$ (Theorem 6(a)).
- By Theorem 24(a) with A and $B = A'$ ($m = 4$, $n = 3$), $t^3 p_{AA'}(t) = t^4 p_{A'A}(t)$, so $p_{AA'} = t\,p_{A'A} = t(t-1)(t-3)(t-4)$ (Proposition 4(a)): the eigenvalues of AA' are 0, 1, 3, 4.
- The vectors $(1,1,1)'$, $(-1,0,1)'$ and $(1,-2,1)'$ are eigenvectors of $A'A$ for 4, 3 and 1:

$$A'A\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}4\\4\\4\end{pmatrix}, \qquad A'A\begin{pmatrix}-1\\0\\1\end{pmatrix} = \begin{pmatrix}-3\\0\\3\end{pmatrix}, \qquad A'A\begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}1\\-2\\1\end{pmatrix}$$

- By Theorem 24(c), applied to A' and A in place of A and B, the map $x \mapsto Ax$ sends them to eigenvectors of AA' for the same eigenvalues:

$$A\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}2\\2\\2\\0\end{pmatrix}, \qquad A\begin{pmatrix}-1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\\2\end{pmatrix}, \qquad A\begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\\0\end{pmatrix}$$

  and indeed $AA'(-1,0,1,2)' = (-3,0,3,6)' = 3(-1,0,1,2)'$.

- The eigenvalue 0 of AA' is simple, so its eigenspace $\mathcal{N}(AA')$ is a line (Theorem 18). Here $A'(1,0,-1,1)' = (1+0+0-1,\ 1+0-1+0,\ 0+0-1+1)' = 0$, so $AA'(1,0,-1,1)' = 0$, and the nonzero vector $(1,0,-1,1)'$ is a basis of the line: $\mathcal{N}(AA') = \text{span}\{(1,0,-1,1)'\}$ (Fact F5).

Practice. The Exercises and Problems of Chapter 7 of the textbook.